PARMIDA DAVARMANESH

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EDUCATION

Massachusetts Institute of Technology Cambridge, MA, USA.

- Ph.D., Computer Science — Expected 2027
  Thesis: Improving open-weight LLMs through steering
  Thesis Advisor: Ashia Wilson
- M.S., Computer Science — 2023
  Thesis: Quantifying 3D genome organization in 4-cell mouse embryos
  Thesis Advisor: Caroline Uhler

University of Michigan Ann Arbor, MI, USA.

- B.S., Honors Mathematics (minor in Computer Science), GPA 3.96/4.0 — 2021

RESEARCH INTERESTS

Machine Learning: Steering, monitoring, prompt optimization, AI alignment, RLVR, LLM post-training, mechanistic interpretability.
Biomedical Sciences: Genomics, precision medicine.

JOURNAL PUBLICATIONS

1. **P. Davarmanesh**, A. Wilson, and A. Radhakrishnan. *Efficient and Accurate Steering of Large Language Models Through Attention-Guided Feature Learning.* Proceedings of the National Academy of Sciences 123(42), Article e2609978123 (2026). Available at PNAS.
2. **P. Davarmanesh**, G. Jabbour, Q. Lin, I. Tenison, C. Stultz, and R. Alam. *Detection of Acute Myocardial Infarction and Ischemia From Lead-I ECG Using Deep Learning.* Circulation 150(Suppl 1), Abstract 4125281 (2024). Available at Circulation.
3. S. Tang, **P. Davarmanesh**, Y. Song, D. Koutra, M. W. Sjoding, and J. Wiens. *Democratizing EHR Analyses with FIDDLE: A Flexible Data-Driven Preprocessing Pipeline for Structured Clinical Data.* Journal of the American Medical Informatics Association 27(12), pages 1921–1934 (2020). Available at JAMIA.

CONFERENCE AND WORKSHOP PUBLICATIONS

1. **P. Davarmanesh**, A. Radhakrishnan, A. Wilson, R. Zhang, J. Tan, J. Pereira, and B. Xia. *Motif-Aware Tokenization of the Genome: Towards Interpretable Modeling of Gene Regulation.* NeurIPS 2025 2nd Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences (2025). Available at OpenReview.
2. **P. Davarmanesh**, Q. Lin, I. Tenison, G. Jabbour, and R. Alam. *Detection of Acute Myocardial Infarction Using Deep Learning on Lead-I ECG Data.* 2024 IEEE 20th International Conference on Body Sensor Networks (BSN), pages 1–4 (2024). Poster presentation. Available at IEEE Xplore. Preprint available at medRxiv.

PREPRINTS AND DATASETS

1. S. Tang, **P. Davarmanesh**, Y. Song, D. Koutra, M. W. Sjoding, and J. Wiens. *MIMIC-III and eICU-CRD: Feature Representation by FIDDLE Preprocessing (version 1.0.0).* PhysioNet (2021). RRID:SCR_007345. Dataset available at PhysioNet.
2. **P. Davarmanesh**, K. Jiang, T. Ou, A. Vysogorets, S. Ivashkevich, M. Kiehn, S. H. Joshi, and N. Malaya. *Automating Artifact Detection in Video Games.* Preprint (2020). Available at arXiv:2011.15103.

SUMMER RESEARCH PROGRAMS

1. **Thermo Fisher Scientific**, Berlin, Germany — 2024
   Macromolecule Classification and Orientation Estimation in Cryo-ET Tomograms
   Mentor: Noushin Hajarolasvadi

2. **Broad Summer Research Program**, Broad Institute of Harvard and MIT 2020
   Blind Three-Dimensional Point Spread Function Estimation from Fluorescent Microscopy Image Stacks Using Generative Adversarial Networks
   Mentor: Mehrtash Babadi

TEACHING

1. Teaching Assistant: *Linear Algebra*, MIT Fall 2026
2. Teaching Assistant: *Mathematical Proof Writing*, University of Michigan 2020 – 2021

HONORS AND AWARDS

1. First Prize, ITA Graduation Day Talks (Information Theory and Applications Workshop) 2026
2. Best Paper Award, MIT LIDS Student Conference 2026
3. Finalist and Audience Choice Award winner ($1,000), MIT $100K Accelerate Entrepreneurship Competition 2025
4. NSF Graduate Research Fellowship Program ($37,000 annually) 2023 – 2026
5. MIT EECS Distinguished Graduate Fellowship ($38,000) 2021
6. Outstanding Undergraduate Researcher Award (honorable mention), Computing Research Association 2020
7. Iranian Scholarship Foundation ($5,000 annually) 2018 – 2020
8. Maryam Mirzakhani Annual Memorial Scholarship ($2,500), Iranian American Scholarship Fund 2017 – 2018

TECHNICAL TALKS

1. INFORMS, San Francisco, CA Nov 2026
2. ML Tea Talk, MIT, Cambridge, MA Oct 2026
3. MIT Statistics and Data Science Conference, Cambridge, MA April 2026
4. Women in Data Science, Cambridge, MA March 2026
5. Information Theory and Applications, San Diego, CA Feb 2026
6. LIDS Student Conference, MIT, Cambridge, MA Jan 2026

LEADERSHIP

MIT Iranian Association Cambridge, MA, USA.

- Founder and Treasurer May 2026 – Present

MIT Volunteer Consulting Group Cambridge, MA, USA.

- Project Leader Sept 2023 – Feb 2024
- Consultant (top 13% of applicants) May – June 2023

Electrical Engineering and Computer Science Graduate Student Association MIT, Cambridge, MA, USA.

- Publicity Chair 2023
- Social Chair 2022